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Small-Covariance Noise-to-State Stability of Stochastic Systems and Its Applications to Stochastic Gradient Dynamics

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Abstract— **This paper studies gradient dynamics subject to additive stochastic noise, which may arise from sources such as stochastic gradient estimation, measurement noise, or stochastic sampling errors. To analyze the robustness of such stochastic gradient systems, the concept of small-covariance noise-to-state stability (NSS) is introduced, along with a Lyapunov-based characterization. Furthermore, the classical Polyak–Łojasiewicz (PL) condition on the objective function is generalized to the $\mathcal{K}$-PL condition via comparison functions, thereby extending its applicability to a broader class of optimization problems. It is shown that the stochastic gradient dynamics exhibit small-covariance NSS if the objective function satisfies the $\mathcal{K}$-PL condition and possesses a globally Lipschitz continuous gradient. This result implies that the trajectories of stochastic gradient dynamics converge to a neighborhood of the optimum with high probability, with the size of the neighborhood determined by the noise covariance. Moreover, if the $\mathcal{K}$-PL condition is strengthened to a $\mathcal{K}_\infty$-PL condition, the dynamics are NSS; whereas if it is weakened to a general positive-definite-PL condition, the dynamics exhibit integral NSS. The results further extend to objectives without globally Lipschitz gradients through appropriate step-size tuning. The proposed framework is further applied to the robustness analysis of policy optimization for the linear quadratic regulator (LQR) and logistic regression.**

I. INTRODUCTION

Optimization lies at the core of many data-driven fields, providing concepts and tools for algorithm design, computational-complexity analysis, and statistical inference. Recent work in learning and optimization largely focuses on gradient-based methods because of their low per-iteration cost and suitability for parallel architectures. Their scalability to large-scale problems has, in turn, motivated careful study of convergence rates—both how to certify a target rate and how to systematically improve it through choices such as step-size policies and momentum. Yet efficiency alone is insufficient: in practice, gradient-based methods operate in noisy environments, where perturbations arise from numerical errors, measurement noise, inexact formulas for gradient computations, and early stopping of embedded routines used to compute gradients (see [1, Ch. 4] and [2, p. 38]). Under such unpredictable noise, the gradient-based algorithms may oscillate around the optimum, converge to a biased limit point, or even diverge. Hence, convergence and robustness are twin, indispensable facets that must be analyzed for gradient-based methods.

Building on the concepts of input-to-state stability (ISS) [3], the authors of [4], [5] analyze continuous-time gradient flows and discrete-time gradient descent under deterministic perturbations. By generalizing the Polyak–Łojasiewicz (PL) condition via comparison functions, they show that the perturbed gradient-flow/gradient-descent dynamics are small-disturbance ISS whenever the objective satisfies the $\mathcal{K}$-PL condition. Specifically, the $\mathcal{K}$-PL condition requires $\|\nabla \mathcal{J}\| \geq \mu(\mathcal{J}(z) - \mathcal{J}^*)$, where $\mathcal{J}$ is the objective, $\mathcal{J}^*$ is the minimum value, and μ is a $\mathcal{K}$-function (continuous, strictly increasing, and vanishing at zero). This framework subsumes the classical PL condition: choosing $\mu(r) = \sqrt{cr}$ with $c > 0$ recovers the classic PL/gradient dominant condition. If μ is strengthened to a $\mathcal{K}_\infty$-function, the perturbed gradient-flow/gradient-descent dynamics are ISS [6]; if μ is relaxed to a positive-definite function, the dynamics are integral ISS [7], [8]. Convergence rates for gradient flows under such generalized PL conditions are analyzed in [9]. Most prior work treats deterministic disturbances. In practice, however, noise is stochastic: objective values and gradients are measured with random errors, and in stochastic optimization the objective is an expectation, so exact gradients are unavailable and must be approximated by random samples. This motivates analyzing gradient methods under random perturbations. Noise-to-state stability (NSS) extends ISS to stochastic systems and is particularly apt when bounding the state by the supremum of unbounded white noise is infeasible [10].

In this paper, we study the robustness of gradient dynamics with additive stochastic perturbations within the NSS framework and introduce small-covariance NSS, which asserts that if the noise covariance is sufficiently small, trajectories enter—and remain in—a neighborhood of the equilibrium with high probability; the neighborhood size depends monotonically (and nonlinearly) on the covariance. Leveraging this notion, we show that if the objective satisfies the $\mathcal{K}$-PL condition, then overdamped Langevin dynamics with time-varying covariance is small-covariance NSS. If the $\mathcal{K}$-PL condition is strengthened to a $\mathcal{K}_\infty$-PL condition (i.e. μ is unbounded), the dynamics are NSS; if it is relaxed to a merely positive-definite function (dropping monotonicity of μ), the dynamics

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are integral NSS [11]. Finally, we illustrate the framework on policy optimization for the linear–quadratic regulator (LQR) and on logistic regression. Due to space limitations, proofs are deferred to the extended version available on arXiv [12].

II. Notations and Facts

A. Notations

In this article, let $\mathbb{R}$ and $\mathbb{R}_+$ denote the set of real numbers and the set of nonnegative real numbers, respectively. Let $\mathbb{R}^n$ denote the n-dimensional Euclidean space. The symbol $\|\cdot\|$ denotes the Euclidean norm of a vector or the spectral norm of a matrix, while $\|\cdot\|_{\mathrm{F}}$ denotes the Frobenius norm of a matrix, where $\|A\|_{\mathrm{F}}^2 = \mathrm{Tr}\left(A^\top A\right)$ for $A \in \mathbb{R}^{n\times m}$. For a measurable and essentially bounded function $x : \mathbb{R}_+ \to \mathbb{R}^n$ (or $X : \mathbb{R}_+ \to \mathbb{R}^{n\times m}$), its essential supremum norm is denoted by $\|x\|_\infty = \operatorname{ess\,sup}_{t\in\mathbb{R}_+}\|x(t)\|$ (or $\|X\|_\infty = \operatorname{ess\,sup}_{t\in\mathbb{R}_+}\|X(t)\|$).

Let $\mathcal{C}(\mathcal{S},\mathbb{R}^n)$, $\mathcal{C}^1(\mathcal{S},\mathbb{R}^n)$ and $\mathcal{C}^2(\mathcal{S},\mathbb{R}^n)$ denote the spaces of continuous, continuously differentiable, and twice continuously differentiable functions from $\mathcal{S} \subset \mathbb{R}^n$ to $\mathbb{R}^n$, respectively. For a function $\mathcal{V} \in \mathcal{C}^2(\mathcal{S},\mathbb{R}^n)$, its gradient and Hessian are denoted by $\nabla\mathcal{V}$ and $\nabla^2\mathcal{V}$, respectively. Let $(\Omega,\mathcal{F},\{\mathcal{F}_t\}_{t\geq 0},\mathbb{P})$ be a complete filtered probability space, where Ω is the sample space, $\mathcal{F}$ is a σ-algebra over Ω, $\{\mathcal{F}_t\}_{t\geq 0}$ is a filtration satisfying the usual conditions (right-continuity and completeness), and $\mathbb{P}$ is a probability measure defined on $(\Omega,\mathcal{F})$.

B. Notions of Comparison Functions

The notions of comparison functions are introduced to facilitate the stability analysis of dynamical systems. A function $\alpha : \mathbb{R}_+ \to \mathbb{R}_+$ is said to be positive definite ($\mathcal{PD}$) if $\alpha(0) = 0$, and $\alpha(r) > 0$ for all $r \neq 0$. A function $\alpha : \mathbb{R}_+ \to \mathbb{R}_+$ is said to be of class $\mathcal{K}$ if it is continuous, strictly increasing, and satisfies $\alpha(0) = 0$. It is said to be of class $\mathcal{K}_\infty$ if $\alpha \in \mathcal{K}$ and $\alpha(r) \to \infty$ as $r \to \infty$. A function α is said to be of class $\mathcal{K}_{[0,d)}$ for some $d > 0$ if it is defined on $[0, d)$, continuous, strictly increasing, and satisfies $\alpha(0) = 0$. A function $\beta : \mathbb{R}_+ \times \mathbb{R}_+ \to \mathbb{R}_+$ is said to be of class $\mathcal{KL}$ if for each $t \geq 0$, the function $\beta(\cdot,t) \in \mathcal{K}$, and for each fixed $r \geq 0$, $\beta(r,\cdot)$ is decreasing and satisfies $\lim_{t\to\infty}\beta(r,t) = 0$.

III. Preliminaries of Noise-to-State Stability

A. Introduction of Nonlinear Stochastic Systems

Consider the following nonlinear stochastic system defined on the probability space $(\Omega,\mathcal{F},\{\mathcal{F}_t\}_{t\geq 0},\mathbb{P})$

$$\mathrm{d}\chi(\omega,t) = f(\chi(\omega,t))\mathrm{d}t + g(\chi(\omega,t))\Sigma(t)\mathrm{d}B(\omega,t), \quad (1)$$

where $\chi : \Omega \times [0,t_{max}) \to \mathcal{S} \subset \mathbb{R}^n$ is an n-dimensional stochastic process; the state space $\mathcal{S} \subset \mathbb{R}^n$ is an open subset diffeomorphic to $\mathbb{R}^n$; the process $B : \Omega \times [0,+\infty) \to \mathbb{R}^m$ denotes a standard m-dimensional Brownian motion; the drift field $f : \mathcal{S} \subset \mathbb{R}^n \to \mathbb{R}^n$ and the diffusion field $g : \mathcal{S} \subset \mathbb{R}^n \to \mathbb{R}^{n\times m}$ are both locally bounded and locally Lipschitz continuous; $f(\chi^*) = 0$ for some equilibrium $\chi^* \in \mathbb{R}^n$, and the matrix-valued function $\Sigma : \mathbb{R}_+ \to \mathbb{R}^{m\times m}$ is Borel measurable and locally essentially bounded. At each time $t \in \mathbb{R}_+$, $\Sigma(t)$ linearly transforms the Brownian noise $\mathrm{d}B(\omega,t)$, resulting in an input $\Sigma(t)\mathrm{d}B(\omega,t)$ with instantaneous covariance $\Sigma(t)\Sigma(t)^\top\mathrm{d}t$. In the following, we refer to $\|\Sigma(t)\Sigma(t)^\top\|$ as the instantaneous noise intensity.

The infinitesimal generator is introduced below to facilitate the stability analysis of the nonlinear stochastic system.

Definition 3.1: The infinitesimal generator $\mathcal{L}$ associated with system (1), acting on a function $\mathcal{V} \in \mathcal{C}^2(\mathcal{S},\mathbb{R}_+)$, is defined as a mapping $\mathcal{L}[\mathcal{V}] : \mathcal{S} \times \mathbb{R}^{m\times m} \to \mathbb{R}$ given by

$$\mathcal{L}[\mathcal{V}](\xi,\Theta) = \langle\nabla\mathcal{V}(\xi), f(\xi)\rangle + \frac{1}{2}\langle g(\xi)\Theta, \nabla^2\mathcal{V}(\xi)g(\xi)\Theta\rangle \quad (2)$$

The infinitesimal generator $\mathcal{L}[\mathcal{V}](\xi,\Theta)$ can be interpreted as the rate of change of the expected value of $\mathcal{V}$ at state ξ under the influence of noise covariance $\Theta\Theta^\top$. It serves as a stochastic counterpart to the Lie derivative.

B. Notions of Noise-to-State Stability

Since system (1) is defined on an open subset, a size function is introduced to facilitate stability analysis within the open subset and to ensure that system trajectories remain inside the domain almost surely.

Definition 3.2: (Size Function) A function $\mathcal{V} : \mathcal{S} \to \mathbb{R}_+$ is a size function for $(\mathcal{S},\chi^*)$ if $\mathcal{V}$ is

1) twice continuously differentiable;
2) positive definite with respect to χ^*, i.e. $\mathcal{V}(\chi^*) = 0$ and $\mathcal{V}(\chi) > 0$ for all $\chi \neq \chi^*$, $\chi \in \mathcal{S}$;
3) coercive, i.e. for any sequence $\{\chi_k\}_{k=0}^\infty$, $\chi_k \to \partial\mathcal{S}$ or $\|\chi_k\| \to \infty$, it holds that $\mathcal{V}(\chi_k) \to \infty$, as $k \to \infty$.

For system (1), the presence of persistent additive stochastic noise—even at equilibrium—prevents trajectories from converging to the equilibrium point in the conventional asymptotic sense. Consequently, the standard notion of stochastic asymptotic stability—requiring that trajectories remain within a ball of radius $\gamma(\|\chi_0\|)$ for some $\gamma \in \mathcal{K}$ with high probability and that $\lim_{t\to\infty}\|\chi(t)\| = 0$ almost surely—no longer applies. Instead, one typically expects the state trajectories to eventually settle into a neighborhood of the equilibrium with high probability, where the size of the neighborhood depends on the noise intensity, quantified by magnitude of the noise covariance $\|\Sigma\Sigma^\top\|_\infty$. The concept of noise-to-state stability (NSS) provides a natural extension of input-to-state stability to stochastic systems, offering a meaningful framework in stochastic scenarios where bounding the state directly by the supremum of unbounded white noise is not feasible.

Definition 3.3: (Noise-to-State Stability in Probability [10]) System (1) is NSS in probability if there exists a size function $\mathcal{V}$ such that for each $\epsilon \in (0,1)$, the following holds

$$\mathbb{P}\big\{\mathcal{V}(\chi(t)) \leq \beta(\mathcal{V}(\chi(0)),t) + \gamma(\|\Sigma\Sigma^\top\|_\infty)\big\} \geq 1-\epsilon \quad (3)$$

for some $\beta \in \mathcal{KL}$, $\gamma \in \mathcal{K}$, all $x(0) \in \mathcal{S}$, and any $t \in \mathbb{R}_+$. By the causality of the dynamical system, the definition remains unchanged if $\|\Sigma\Sigma^\top\|_\infty$ is replaced by $\operatorname{ess\,sup}_{\tau\in[0,t]}\|\Sigma(\tau)\Sigma(\tau)^\top\|$. The aforementioned notion of NSS requires that trajectories remain bounded with high probability for arbitrarily large noise intensity (i.e.

$\|\Sigma\Sigma^\top\|_\infty$). However, in practice, many stochastic systems may diverge under large noise covariance and only exhibit NSS behavior when the noise covariance is sufficiently small. This observation motivates the introduction of small-covariance NSS, which relaxes the classical NSS condition by characterizing stability properties that hold only under noise with small covariance.

Definition 3.4: (Small-Covariance Noise-to-State Stability in Probability) System (1) is small-covariance noise-to-state stable (scNSS) in probability if there exists a size function $\mathcal{V}$ and a constant $d > 0$, such that for each $\epsilon \in (0, 1)$, the following holds

$$\mathbb{P}\big\{\mathcal{V}(\chi(t)) \leq \beta(\mathcal{V}(\chi(0)), t) + \gamma(\|\Sigma\Sigma^\top\|_\infty)\big\} \geq 1 - \epsilon \quad (4)$$

for some $\beta \in \mathcal{KL}$, $\gamma \in \mathcal{K}_{[0,d)}$, all noise covariances bounded by d (i.e. $\|\Sigma\Sigma^\top\|_\infty < d$), all $x(0) \in \mathcal{S}$, and any $t \in \mathbb{R}_+$.

Analogous to integral input-to-state stability (iISS) in deterministic systems, the state trajectories can be bounded by an energy-like functional of the noise intensity. This observation motivates the notion of integral noise-to-state stability (iNSS) for stochastic systems.

Definition 3.5: (Integral Noise-to-State Stability in Probability [11]) System (1) is iNSS in probability if there exists a size function $\mathcal{V}$ such that for each $\epsilon \in (0, 1)$, the following holds

$$\begin{aligned}\mathbb{P}\Big\{&\mathcal{V}(\chi(t)) \leq \beta(\mathcal{V}(\chi(0)), t) \\ &+ \int_0^t \gamma(\|\Sigma(\tau)\Sigma(\tau)^\top\|)\mathrm{d}\tau\Big\} \geq 1 - \epsilon\end{aligned} \quad (5)$$

for some $\beta \in \mathcal{KL}$, $\gamma \in \mathcal{K}$, all $x(0) \in \mathcal{S}$, and any $t \in \mathbb{R}_+$.

C. Lyapunov Functions of Noise-to-State Stability

Lyapunov functions offer a tractable approach to determine whether a stochastic system satisfies NSS. The notion of Lyapunov function for NSS is introduced below.

Definition 3.6: (NSS-Lyapunov Function) A function $\mathcal{V} : \mathcal{S} \to \mathbb{R}_+$ is an NSS-Lyapunov function if

1) $\mathcal{V} : \mathcal{S} \to \mathbb{R}_+$ is a size function for $(\mathcal{S}, \chi^*)$;
2) there exist $\alpha \in \mathcal{K}_\infty$ and $\gamma \in \mathcal{K}$ such that

$$\mathcal{L}[\mathcal{V}](\xi, \Theta) \leq -\alpha(\mathcal{V}(\xi)) + \gamma(\|\Theta\Theta^\top\|) \quad (6)$$

for all $\xi \in \mathcal{S}$ and all $\Theta \in \mathbb{R}^{m\times m}$.

A scNSS-Lyapunov function can be derived from an NSS-Lyapunov function by relaxing the $\mathcal{K}_\infty$-function α in Definition 3.6 to a general $\mathcal{K}$-function, and replacing γ with a $\mathcal{K}_{[0,d)}$-function.

Definition 3.7: (scNSS-Lyapunov Function) A size function $\mathcal{V} : \mathcal{S} \to \mathbb{R}_+$ is a scNSS-Lyapunov function if there exist $\alpha \in \mathcal{K}$ and $\gamma \in \mathcal{K}_{[0,d)}$ such that (6) holds for all $\xi \in \mathcal{S}$ and all $\Theta \in \mathbb{R}^{m\times m}$ satisfying $\|\Theta\Theta^\top\| < d$.

By comparing the NSS-Lyapunov function in Definition 3.6 with the scNSS-Lyapunov function in Definition 3.7, we observe that the generator term $\mathcal{L}[\mathcal{V}](\xi, \Theta)$ may become unbounded as $\|\Theta\Theta^\top\| \to d$ in the scNSS case, whereas it remains bounded for the NSS-Lyapunov function. The dissipative-type Lyapunov function for iNSS is introduced below.

Definition 3.8: (iNSS-Lyapunov Function) A size function $\mathcal{V} : \mathcal{S} \to \mathbb{R}_+$ is an iNSS-Lyapunov function if there exist $\alpha \in \mathcal{PD}$ and $\gamma \in \mathcal{K}$ such that (6) holds for all $\xi \in \mathcal{S}$ and all $\Theta \in \mathbb{R}^{m\times m}$.

The existence of a scNSS-Lyapunov function is sufficient to establish the scNSS property of a stochastic system. We are now ready to state the main result of this section.

Theorem 3.1: System (1) is scNSS if there exists a scNSS-Lyapunov function for it.

The existence of an NSS-Lyapunov function implies that system (1) is NSS. This result can be viewed as a special case of the scNSS property.

Theorem 3.2: System (1) is NSS if there exists a NSS-Lyapunov function for it.

As shown in [11, Theorem 2], the existence of an iNSS-Lyapunov function guarantees that system (1) is iNSS.

Theorem 3.3 ([11]): System (1) is iNSS if there exists an iNSS-Lyapunov function for it.

In the next section, we apply the developed notions of NSS to analyze the robustness of a specific class of stochastic dynamics: stochastic gradient dynamics.

IV. NSS for Stochastic Gradient Dynamics

In this section, we apply different notions of NSS to analyze gradient flows under stochastic noise with time-varying covariance.

A. Robustness of Overdamped Langevin Diffusion

Gradient flows are efficient for solving the optimization problem

$$\min_{z\in\mathcal{Z}} \mathcal{J}(z) \quad (7)$$

where $z \in \mathbb{R}^n$ is the decision variable and $\mathcal{Z} \subset \mathbb{R}^n$ is a feasible set that is diffeomorphic to $\mathbb{R}^n$. By continuously updating the decision variable in the direction of gradient descent

$$\mathrm{d}z(t) = -\nabla\mathcal{J}(z(t))\mathrm{d}t, \quad (8)$$

the algorithm drives the decision variables toward the set of critical points $\mathcal{Z}_c = \{z \in \mathcal{Z} | \nabla\mathcal{J}(z) = 0\}$ under mild regularity conditions:

Assumption 4.1: The objective function $\mathcal{J}$ is twice continuously differentiable, bounded below, and coercive on the set $\mathcal{Z}$.

Moreover, in the deterministic case, for many examples Assumption 4.1 helps rule out convergence to strict saddle points or local maxima [13], [14], thereby ensuring convergence to a local minimum. To guarantee convergence to the global minimum $\mathcal{J}^*$ of problem (7), additional assumptions are required–most notably, convexity of $\mathcal{J}$. Another sufficient condition, which ensures exponential convergence to the global minimum, was proposed in [15], [16]:

Definition 4.1: (Polyak-Łojasiewicz (PL) Condition) The function $\mathcal{J}$ satisfies the Polyak-Łojasiewicz (PL) condition if there exists $c > 0$ such that

$$\|\nabla \mathcal{J}(z)\| \geq \mu(\mathcal{J}(z) - \mathcal{J}^*), \tag{9}$$

where $\mu(r) = \sqrt{cr}$ for all $r \in \mathbb{R}_+$.

While convergence is a key consideration, robustness is another critical factor that determines the efficacy of a gradient algorithm. In the presence of noise, the algorithm should still converge to a solution that is close to the optimum. As noted by [1], many disturbances affecting gradient descent methods can be modeled as random noise. In such cases, the gradient dynamics can be represented by a stochastic differential equation, often referred to as the *overdamped Langevin diffusion*:

$$\mathrm{d}z(t) = -\nabla \mathcal{J}(z(t))\mathrm{d}t + G(z(t))\Sigma(t)\mathrm{d}B(t) \tag{10}$$

where B is a standard, independent n-dimensional Brownian motion, $G : \mathcal{Z} \to \mathbb{R}^{n\times n}$ is locally Lipschitz continuous and locally bounded, and $\Sigma : \mathbb{R}_+ \to \mathbb{R}^{n\times n}$ is Borel measurable and locally essentially bounded, which characterizes the intensity of the stochastic perturbations. Under persistent stochastic noise, the gradient dynamics can never remain exactly at a critical point where $\nabla \mathcal{J}(z) = 0$. Instead, we aim for the algorithm to remain within a neighborhood of the optimum with high probability. The size of this neighborhood is governed by the intensity of the noise, characterized by $\|\Sigma\Sigma^\top\|_\infty$. Formally, this requirement can be expressed by demanding that the stochastic gradient dynamics exhibit NSS.

We aim to establish a connection between the PL condition and the NSS of stochastic gradient dynamics. The classical PL condition requires μ to be the square-root function, which can be restrictive and limit its applicability in broader contexts where such a strict condition may not hold. To address this limitation, we introduce relaxed versions of the PL condition by employing comparison functions.

Definition 4.2: ($\mathcal{K}_\infty$-PL Condition) The function $\mathcal{J}$ satisfies the $\mathcal{K}_\infty$-PL condition if there exists a $\mu \in \mathcal{K}_\infty$ such that (9) holds.

The $\mathcal{K}_\infty$-PL condition can be further relaxed by removing the unboundedness requirement of the $\mathcal{K}_\infty$-function.

Definition 4.3: ($\mathcal{K}$-PL Condition) The function $\mathcal{J}$ satisfies the $\mathcal{K}$-PL condition if there exists a $\mu \in \mathcal{K}$ such that (9) holds.

The $\mathcal{K}$-PL condition can be further relaxed by removing the monotonicity requirement of the $\mathcal{K}$-function.

Definition 4.4: ($\mathcal{PD}$-PL Condition) The function $\mathcal{J}$ satisfies the $\mathcal{PD}$-PL condition if there exists a $\mu \in \mathcal{PD}$ such that (9) holds.

1) Robustness Analysis with Global Lipchitz Condition: The following main theorem establishes the connection between NSS and the different variants of the PL conditions by assuming that $\nabla\mathcal{J}$ is globally Lipchitz continuous.

Theorem 4.1: Let the objective function $\mathcal{J}$ have a global L-Lipschitz continuous gradient, and let $G(z)$ be globally bounded, i.e., $\|G\|_{\mathrm{F}} \leq K_G$. Then:

1) If the function $\mathcal{J}$ satisfies the $\mathcal{K}_\infty$-PL condition, the overdamped Langevin diffusion (10) is NSS;
2) If the function $\mathcal{J}$ satisfies the $\mathcal{K}$-PL condition, the overdamped Langevin diffusion (10) is scNSS.
3) If the function $\mathcal{J}$ satisfies the $\mathcal{PD}$-PL condition, the overdamped Langevin diffusion (10) is iNSS.

2) Robustness Analysis without Global Lipchitz Condition: In the preceding discussion, the global Lipschitz continuity of $\nabla\mathcal{J}(z)$ is required to establish the connection between various versions of NSS and the generalized PL conditions. However, this global assumption may limit practical applicability. To address this, we relax the global Lipschitz requirement by appropriately tuning the learning rate in the gradient dynamics. Specifically, we allow for a state-dependent learning rate $\eta(\mathcal{J}(z)) > 0$ in the dynamics:

$$\mathrm{d}z(t) = -\eta(\mathcal{J}(z(t)))\nabla \mathcal{J}(z(t))\mathrm{d}t + G(z(t))\Sigma(t)\mathrm{d}B(t) \tag{11}$$

Since $\mathcal{J}$ is twice continuously differentiable, its gradient is locally Lipschitz and its Hessian is locally bounded. To facilitate the robustness analysis, define the sublevel set

$$\mathcal{Z}_h = \{z \in \mathcal{Z} | \, \mathcal{J}(z) - \mathcal{J}^* \leq h\}. \tag{12}$$

Note that for any $z \in \mathcal{Z}$, we have $z \in \mathcal{Z}_{\mathcal{J}(z)-\mathcal{J}^*}$. Over the sublevel set, define

$$\bar{L}(h) = \frac{1}{2}\max_{z\in\mathcal{Z}_h} \|\nabla^2\mathcal{J}(z)\|\|G(z)\|_{\mathrm{F}}^2 \tag{13}$$

which is a continuous and nondecreasing function of h. Then, the function

$$\tilde{L}(h) = \bar{L}(h) - \bar{L}(0) \tag{14}$$

can be assumed to be of $\mathcal{K}_\infty$, since otherwise it can be modify by adding an identity function, ensuring that $\tilde{L}(h) + h \in \mathcal{K}_\infty$.

Theorem 4.2: Suppose $\mathcal{J}(z)$ satisfies the $\mathcal{K}$-PL condition. Then,

1) if $\lim_{h\to\infty} \frac{\tilde{L}(h)}{\eta(h)\mu(h)^2} = d_2 > 0$, the overdamped Langevin diffusion in (11) is scNSS;
2) if $\lim_{h\to\infty} \frac{\tilde{L}(h)}{\eta(h)\mu(h)^2} = 0$, the overdamped Langevin diffusion in (11) is NSS.

As a concrete example, one may choose $\eta(h) = \tilde{L}(h)$ such that $\lim_{h\to\infty} \frac{\tilde{L}(h)}{\eta(h)\mu(h)^2} = \lim_{h\to\infty} \frac{1}{\mu(h)^2} = d_2$. Then, the stochastic gradient dynamics is scNSS. If $\eta(h) = h\tilde{L}(h)$, $\lim_{h\to\infty} \frac{\tilde{L}(h)}{\eta(h)\mu(h)^2} = 0$ and the stochastic gradient dynamics is NSS.

V. Applications

In this section, we apply the notion of NSS to analyze the robustness of the stochastic gradient dynamics for two important problem classes: policy optimization for LQR and logistic regression.

A. Policy Optimization for Linear Quadratic Regulator

We study the policy optimization algorithm for the following linear system:

$$\dot{x}(t) = Ax(t) + Fu(t),\, x(0) = x_0, \tag{15}$$

where $A \in \mathbb{R}^{n\times n}$ and $F \in \mathbb{R}^{n\times m}$ are constant system matrices; $x(t) \in \mathbb{R}^n$ and $u(t) \in \mathbb{R}^m$ denote the states and control inputs, respectively. The performance index of the system is defined as

$$\mathcal{J}_1(x_0, u) = \int_0^\infty x(t)^\top Qx(t) + u(t)^\top Ru(t)\mathrm{d}t \tag{16}$$

where $Q \in \mathbb{S}^n_{++}$ and $R \in \mathbb{S}^n_{++}$. Under the assumption that the pair (A, F) is controllable, the optimal control for (16) is given by

$$u^*(t) = -R^{-1}F^\top P^*x(t) = -K^*x(t) \tag{17}$$

where $P^* \succ 0$ is the solution of the Riccati equation

$$A^\top P^* + P^*A - P^*FR^{-1}F^\top P^* + Q = 0. \tag{18}$$

Policy optimization seeks the best feedback control K^* over the admissible set

$$\mathcal{G} = \{K \in \mathbb{R}^{m\times n} | \, (A - FK) \text{ is Hurwitz}\}, \tag{19}$$

by solving

$$\min_{K\in\mathcal{G}} \mathcal{J}_2(K) = \mathbb{E}_{x_0\sim\mathcal{N}(0,I_n)}\mathcal{J}_1(x_0, -Kx) = \mathrm{Tr}\left(P_K\right), \tag{20}$$

where $P_K \in \mathbb{S}^n_{++}$ is the solution of the Lyapunov equation

$$(A - FK)^\top P_K + P_K(A - FK) + Q + K^\top RK = 0. \tag{21}$$

The gradient of the objective function $\mathcal{J}_2$ is given by

$$\nabla\mathcal{J}_2(K) = 2(RK - F^\top P_K)Y_K \tag{22}$$

where $Y_K \in \mathbb{S}^n_{++}$ is the solution of the Lyapunov equation

$$(A - FK)Y_K + Y_K(A - FK)^\top + I_n = 0. \tag{23}$$

Let Y^* denote the solution of (23) corresponding to the optimal feedback gain K^*, and $\mathcal{J}_2^* = \mathcal{J}_2(K^*)$. The following lemma shows the local Lipchitz continuity of $\nabla\mathcal{J}_2$.

Lemma 5.1 (Lemma 5.3 in [5]): The gradient $\nabla\mathcal{J}_2(K)$ is $\bar{L}_3(h)$-Lipschitz continuous over the sublevel set $\mathcal{G}_h = \{K \in \mathcal{G} | \mathcal{J}_2(K) - \mathcal{J}_2^* \le h\}$, with

$$\begin{aligned}\bar{L}_3(h) = &\frac{2\|R\|}{\lambda_{\min}(Q)}(\mathcal{J}_2^* + h) + \frac{8a_2\|F\|\|R\|}{\lambda_{\min}(Q)^2}(\mathcal{J}_2^* + h)^{\frac{5}{2}} \\ &+ \frac{8\|F\|(a_1\|R\| + \|F\|)}{\lambda_{\min}(Q)^2}(\mathcal{J}_2^* + h)^3.\end{aligned} \tag{24}$$

where

$$a_1 = \frac{2\|F\|}{\lambda_{\min}(R)},\, a_2 = \left(\frac{2\|A\|}{\lambda_{\min}(R)}\right)^{1/2} \tag{25}$$

The following lemma shows that $\mathcal{J}_2(K) - \mathcal{J}_2^*$ can serve as a size function over $\mathcal{G}$.

Lemma 5.2 (Proposition 3.2 and Lemma 3.3 in [17]): The function $\mathcal{J}_2(K)$ is coercive and analytic over $\mathcal{G}$.

The following lemma shows that the objective function $\mathcal{J}_2$ satisfies the $\mathcal{K}$-PL condition.

Lemma 5.3 ($\mathcal{K}$-PL condition of LQR [4]): The objective function $\mathcal{J}_2(K)$ satisfies the $\mathcal{K}$-PL condition, i.e.,

$$\|\nabla\mathcal{J}_2(K)\|_F \ge \mu_5(\mathcal{J}_2(K) - \mathcal{J}_2^*)\,, \quad \forall K \in \mathcal{G},$$

where

$$\mu_5(h) = \frac{h}{b_1 h + b_2}, \qquad h \ge 0, \tag{26}$$

and b_1 and b_2 are positive constants.

The overdamped Langevin diffusion for solving the policy optimization problem (20) is given by

$$\begin{aligned}\mathrm{d}K(s) = &-2\eta(\mathcal{J}(K(s)) - \mathcal{J}^*) \\ &(RK(s) - F^\top P(s))Y(s)\mathrm{d}s + \Sigma_1(s)\mathrm{d}W(s)\end{aligned} \tag{27}$$

where $W(s) \in \mathbb{R}^{m\times n}$ is a standard, independent Brownian motion, $P(s) = P_{K(s)}$, $Y(s) = Y_{K(s)}$, and $\Sigma_1 : \mathbb{R}_+ \to \mathbb{R}^{m\times m}$ is Borel measurable and locally essentially bounded, which characterizes the intensity of the stochastic perturbations. Since the gradient of $\mathcal{J}_2$ is not globally Lipschitz continuous, the NSS properties of (27) can be established by Theorem 4.2.

Theorem 5.1: The overdamped Langevin diffusion in (27) is NSS if $\lim_{h\to\infty} \frac{h^3}{\eta(h)} = 0$, and it is scNSS if $0 < \lim_{h\to\infty} \frac{h^3}{\eta(h)} < \infty$.

B. Logistic Regression

Suppose we are given a dataset of N samples $\{(x_i, y_i)\}_{i=1}^N$, where each feature vector $x_i \in \mathbb{R}^n$ and the corresponding label $y_i \in \{0, 1\}$. In logistic regression, the probability that a sample with feature vector x belongs to class $y = 1$ is modeled as

$$p(x;\theta) = \mathbb{P}(y = 1|x;\theta) = \sigma(\theta^\top x) = \frac{1}{1 + \exp(-\theta^\top x)}, \tag{28}$$

where $\theta \in \mathbb{R}^n$ denotes the parameter vector to be learned. The model parameters are obtained by minimizing the empirical negative log-likelihood (binary cross-entropy) loss:

$$\begin{aligned}&\min_{\theta\in\mathbb{R}^n} \mathcal{J}_3(\theta) = \frac{1}{N}\sum_{i=1}^N \ell_i(\theta) \\ &= -\frac{1}{N}\sum_{i=1}^N y_i \log(p_i(\theta)) + (1 - y_i)\log(1 - p_i(\theta)),\end{aligned} \tag{29}$$

where $\ell_i(\theta)$ is the per-sample loss and $p_i(\theta) = p(x_i;\theta)$. Let $X = [x_1, x_2, \cdots, x_n] \in \mathbb{R}^{n\times N}$ denote the data matrix. The gradient and Hessian of the logistic loss are given by

$$\nabla\mathcal{J}_3(\theta) = \frac{1}{N}\sum_{i=1}^N (p_i(\theta) - y_i)x_i, \tag{30a}$$

$$\nabla^2\mathcal{J}_3(\theta) = \frac{1}{N}\sum_{i=1}^N p_i(\theta)(1 - p_i(\theta))x_i x_i^\top = \frac{1}{N}X\Lambda(\theta)X^\top, \tag{30b}$$

where $\Lambda(\theta) = \mathrm{diag}(p_i(\theta)(1 - p_i(\theta)))$. The following standard assumptions are imposed throughout the analysis.

Assumption 5.1: The dataset is sufficiently rich so that the data matrix $X \in \mathbb{R}^{n \times N}$ has full row rank, i.e., $\mathrm{rank}(X) = n$.

Assumption 5.2: The dataset is *nonseparable*. That is, there does not exist a nonzero vector $\vec{\theta} \in \mathbb{R}^n$, such that $\vec{\theta}^\top x_i \geq 0$ for all i with $y_i = 1$ and $\vec{\theta}^\top x_i \leq 0$ for all i with $y_i = 0$.

1) Properties of Logistic Loss: The following lemmas establish key properties of $\mathcal{J}_3$. In particular, under Assumption 5.2, the objective function is shown to be coercive.

Lemma 5.4: Under Assumption 5.2, the objective function $\mathcal{J}_3$ is coercive over $\mathbb{R}^n$.

We now present a lemma that demonstrates the convexity and smoothness properties of $\mathcal{J}_3$.

Lemma 5.5: Under Assumption 5.1, the objective function $\mathcal{J}_3$ is strictly convex, and its gradient is globally $\frac{1}{4N}\|XX^\top\|$-Lipschitz continuous.

Since $\mathcal{J}_3$ is coercive over $\mathbb{R}^n$ and strictly convex, there exists a unique minimizer θ^*. The following lemma lays the foundation for establishing the scNSS property of the gradient-based optimization algorithms used to solve logistic regression.

Lemma 5.6: Under Assumptions 5.2, the objective function $\mathcal{J}_3$ satisfies the $\mathcal{K}$-PL condition, i.e.,

$$\|\nabla \mathcal{J}_3(\theta)\| \;\geq\; \mu_6\left(\mathcal{J}_3(\theta) - \mathcal{J}_3^*\right), \tag{31}$$

where μ_6 is a $\mathcal{K}$-function.

Remark 5.1: Even though the logistic loss $\mathcal{J}_3$ is strictly convex (Lemma 5.5), it cannot satisfy the $\mathcal{K}_\infty$-PL condition. Indeed, since $|p_i(\theta) - y_i| \leq 1$, we have

$$\|\nabla \mathcal{J}_3(\theta)\| \;\leq\; \|X\|/\sqrt{N}.$$

On the other hand, by Lemma 5.4, $\mathcal{J}_3(\theta) \to \infty$ as $\|\theta\| \to \infty$. Hence, it is impossible to find a $\mathcal{K}_\infty$-function μ such that

$$\|\nabla \mathcal{J}_3(\theta)\| \;\geq\; \mu(\mathcal{J}_3(\theta) - \mathcal{J}_3^*). \tag{32}$$

In addition, since $\lim_{\|\theta\|\to\infty} p_i(\theta)(1 - p_i(\theta)) = 0$, it follows from (30b) that $\lim_{\|\theta\|\to\infty} \nabla^2 \mathcal{J}_3(\theta) = 0$. Hence, the logistic loss is not globally strongly convex either.

2) Robustness Analysis: Under stochastic perturbations, the first-order gradient flow for solving logistic regression (29) can be represented by the following overdamped Langevin diffusion:

$$\mathrm{d}\theta(s) \;=\; -\nabla \mathcal{J}_3\big(\theta(s)\big)\,\mathrm{d}s \;+\; \Sigma(s)\,\mathrm{d}B(s), \tag{33}$$

where $B(s)$ denotes a standard Brownian motion and $\Sigma(s)$ characterizes the covariance structure of the noise. As a direct application of the results developed in Section IV, it follows that the Langevin diffusion in (33) is scNSS.

Theorem 5.2: Under Assumptions 5.1 and 5.2, the overdamped Langevin diffusion in (33) is scNSS.

VI. Conclusions

In this paper, we introduced a new notion—small-covariance NSS—together with a Lyapunov condition for its verification. Small-covariance NSS guarantees that when the covariance of stochastic noise is sufficiently small, system trajectories eventually enter and remain within a neighborhood of the equilibrium, whose size depends (generally nonlinearly) on the noise covariance. Under this framework, we analyzed stochastic gradient dynamics and showed that if the objective function satisfies a generalized $\mathcal{K}$-PL condition, the dynamics are small-covariance NSS. Moreover, strengthening the $\mathcal{K}$-PL condition to a $\mathcal{K}_\infty$-PL condition yields NSS, while weakening it to a positive-definite condition leads to integral NSS. The framework was applied to policy optimization for LQR and logistic regression, showing that the corresponding stochastic gradient dynamics are small-covariance NSS.

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